Difference Between Primary Key and Foreign Key

Here's a comprehensive comparison:

---

1. Quick Comparison Table

Feature	Primary Key	Foreign Key
Purpose	Uniquely identifies each record in a table	Establishes a relationship between two tables
Uniqueness	Must be unique for every row	Can have duplicate values
NULL Values	Cannot be NULL	Can be NULL (unless NOT NULL is specified)
Number per Table	Only ONE primary key per table	Can have multiple foreign keys per table
References	Does not reference any other table	References the PRIMARY KEY of another table
Parent-Child	Acts as the parent	Acts as the child
Automatic Index	Yes, creates a clustered index by default	No, but indexing is recommended for performance
Data Modification	Cannot be modified if referenced by a foreign key	Can be modified/deleted with constraints (CASCADE, RESTRICT, etc.)
Scope	Within the same table	Across multiple tables
Database Constraint	Entity integrity	Referential integrity

---

2. Visual Representation

text

```
┌──────────────────────┐          ┌──────────────────────────┐
│      DEPARTMENT      │          │         EMPLOYEE         │
│    (Parent Table)    │          │      (Child Table)       │
├──────────────────────┤          ├──────────────────────────┤
│  dept_id (PK) ───────┼──────────┼── dept_id (FK)          │
│  dept_name           │          │  emp_id (PK)             │
│  location            │          │  emp_name                │
└──────────────────────┘          │  salary                  │
                                  └──────────────────────────┘


        ▲                              ▲
        │                              │
  PRIMARY KEY                    FOREIGN KEY
  - Unique                        - Can have duplicates
  - Not NULL                      - Can be NULL
  - Only one per table            - Many per table
  - Identifies record             - Links to parent
```

---

3. Detailed Examples

Example Tables:

sql

```
-- PARENT TABLE
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,        -- PRIMARY KEY
    dept_name VARCHAR(50) NOT NULL
);

-- CHILD TABLE
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,          -- PRIMARY KEY
    emp_name VARCHAR(100),
    dept_id INT,                     -- FOREIGN KEY
```

```
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
```

Insertion Behavior:

sql

```
-- Insert into Department (Parent) - Works
INSERT INTO Department VALUES (1, 'IT');
INSERT INTO Department VALUES (2, 'HR');

-- Insert into Employee (Child) - Works (dept_id 1 exists)
INSERT INTO Employee VALUES (101, 'John', 1);

-- Insert into Employee - Fails (dept_id 99 doesn't exist)
INSERT INTO Employee VALUES (102, 'Jane', 99);  -- ERROR!

-- Insert into Employee - Works (NULL allowed unless specified)
INSERT INTO Employee VALUES (103, 'Bob', NULL);
```

---

4. Key Differences Explained

4.1 Uniqueness

Primary Key	**Foreign Key**
Each value must be unique	Values can repeat
`sql |` sql	
dept_id: 1, 2, 3 (OK)	dept_id: 1, 2, 2, 1 (OK)
dept_id: 1, 1, 2 (ERROR)	dept_id: 3, 4, 4 (OK)
`|`	

4.2 NULL Values

Primary Key	**Foreign Key**
Cannot be NULL	Can be NULL

`sql |`sql	
INSERT INTO Department	INSERT INTO Employee
VALUES (NULL, 'IT');	VALUES (101, 'John', NULL);
-- ERROR!	-- OK if NOT NULL not set
`|`	

4.3 Data Integrity Rules

Primary Key ensures:

- No duplicate rows
- Every row can be uniquely identified

Foreign Key ensures:

- Referential integrity - no orphan records
- Data consistency across related tables

sql

```
-- DELETE RESTRICT (default behavior)
DELETE FROM Department WHERE dept_id = 1;
-- ERROR! Cannot delete because employees exist

-- DELETE CASCADE
DELETE FROM Department WHERE dept_id = 1;
-- OK! Also deletes all employees with dept_id=1

-- DELETE SET NULL
DELETE FROM Department WHERE dept_id = 1;
-- OK! Sets dept_id to NULL for affected employees
```

---

5. Analogy to Understand

Think of a Primary Key as a Government ID (Aadhaar/PAN):

- Unique to each person
- Never changes
- Cannot be blank

Think of a Foreign Key as a Company ID Card:

- References your Government ID
- Multiple people can work at the same company (duplicates allowed)
- You might not have one if you're unemployed (NULL allowed)

text

```
Government ID (Primary Key) → Unique per citizen
         ↓
Company ID (Foreign Key) → Links to Government ID
```

---

6. Summary: Key Points to Remember

Aspect	Primary Key	Foreign Key
Simple Definition	Uniquely identifies a row	Links two tables together
Rules	UNIQUE + NOT NULL	Can be duplicate + NULL
Count	1 per table	Multiple per table
Purpose	Entity integrity	Referential integrity
Example	`emp_id` in Employee table	`dept_id` in Employee table (references Department.dept_id)
Independence	Standalone	Depends on primary key of another table

---

7. Quick Quiz

Which of these is valid?

1. `emp_id` in Employee table can be duplicated → ❌ (Primary Key must be unique)
2. `dept_id` in Employee table can be NULL → ✅ (Foreign Key can be NULL)
3. A table can have 3 primary keys → ❌ (Only 1 primary key per table)
4. A table can have 3 foreign keys → ✅ (Multiple foreign keys allowed)
5. Foreign key must reference a primary key → ✅ (Always references PK of another table)